

Final Report — Singapore Connected

CM1040 Web Development Coursework

Student:	VARUN.G
Student ID:	250626847
Programme:	BSc Computer Science (Physical Computing & IoT)
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A. Background Research

Singapore was chosen for this project due to its remarkable transformation into a global digital powerhouse. In just four decades, Singapore built one of the world's most advanced digital infrastructures, with 92% internet penetration and 10 Gbps broadband speeds as of 2024.

Project Scope: The project traces Singapore's digital journey from 1981 (establishment of the National Computer Board, which began the nation's computerisation) to 2026 (Smart Nation 2.0). The first internet connection occurred in 1987 when NUS connected to BITNET — this marks the beginning of Singapore's connected era. The 1981–1986 period is included to provide context on the foundational government initiatives that enabled the internet's arrival.

Research focused on key events during the student's lifetime (1981–2026). Sources prioritised were primary/official sources (GovTech Singapore, IMDA, Smart Nation Singapore, NUS, PMO) and secondary sources (academic publications and reputable technology sources).

Key Milestones Verified

- 1981 – National Computer Board Established – Began Singapore's digital transformation
- 1987 – NUS Connects to BITNET – First permanent BITNET node in SE Asia
- 1990 – NUS First Internet Connection – Singapore joined global Internet community
- 1991 – NUSNET Launched – Advanced academic network in Asia
- 1991 – TechNet Established – Singapore's first ISP
- 1992 – IT2000 Masterplan Adopted – First national IT blueprint
- 1994 – SingNet Launched – First commercial ISP for the public
- 1997 – Singapore ONE Launched – World's first nationwide broadband
- 2003 – Singpass Launched – National digital identity system
- 2006 – iN2015 Masterplan Launched – S\$3.2 billion, 10-year plan
- 2006 – Wireless@SG Launched – First nationwide free Wi-Fi
- 2010 – Next Generation NBN Launched – All-fibre network, 1 Gbps speeds
- 2014 – Smart Nation Vision Launched – National digital transformation strategy
- 2016 – GovTech Formed – Government digital transformation agency
- 2017 – SNDGO Established – Smart Nation strategy coordination
- 2017 – PayNow Launched – Instant peer-to-peer fund transfer
- 2019 – National AI Strategy Unveiled – Positioned Singapore as AI leader
- 2020 – 5G Licences Awarded – Nationwide 5G rollout began
- 2024 – Smart Nation 2.0 Launched – Human-centric digital future

All 22 verified milestones are documented in data/timeline.json with full source citations.

B. Planning

Page Architecture

- Home – Singapore Connected – Overview – Introduction, stats, timeline preview
- Origins – From BITNET to Broadband – 1981–1996 – Academic networks, first ISPs
- Growth – Broadband & Beyond – 1997–2009 – Singapore ONE, wireless, Singpass
- Smart Nation – Building a Smart Nation – 2010–2026 – Fibre, Smart Nation, 5G, AI
- Timeline – Interactive Timeline – 1981–2026 – All 22 milestones in one view

Technology Stack

- Markup – HTML5 – Semantic structure
- Styling – CSS3 – Red & white theme, responsive
- Logic – JavaScript (ES6) – Dynamic rendering, validation
- Data – JSON – Timeline, milestones, statistics
- Template Engine – Custom JavaScript – Converts JSON to HTML
- Hosting – GitHub Pages – Free static deployment

Design System (Singapore Flag Colors)

- Background – White – #FFFFFF
- Primary Text – Dark Red/Black – #1A0000
- Primary Accent – Bright Red – #E02A3E
- Secondary Accent – Dark Red – #8B0000
- Subtle Background – Very Light Red – #FFF5F5

Typography: Playfair Display for headings, Inter for body text. Glassmorphism used in hero sections.

C. Development Process: Prototype Designs

Wireframes were created using Visily for all 5 pages, showing both desktop and mobile layouts, focusing on responsiveness (mobile-first with breakpoints at 768px and 1024px), accessibility (semantic structure, ARIA labels), and aesthetics (Singapore flag colors).

User Feedback

The website was presented to 3 users via Google Form:

- User 1 (Desktop) – "Text could be more readable on hero" – Added glassmorphism overlay
- User 2 (Mobile) – "Navigation menu is easy to use" – Retained hamburger menu
- User 3 (Desktop) – "Timeline events are well organised" – No changes needed

Feedback Form: <https://forms.gle/PJz7v57D9GJUCw4A>

Design Revisions

Based on feedback: 1) Added glassmorphism for better text readability, 2) Adjusted text colours for WCAG AA compliance, 3) Enhanced touch targets to 44px or more for mobile.

User Testing Results

The website prototype was presented to 5 users via Google Form. Users rated navigation ease on a scale of 1–5.

User	Device	Navigation Rating	Key Feedback
User 1	Desktop	5/5	About the internet — clear understanding
User 2	Mobile	5/5	Make homepage more visually engaging — add more visuals
User 3	Mobile	5/5	Good design, not too complicated, no changes required
User 4	Desktop	5/5	Clean design
User 5	Mobile	5/5	Not clear what exactly the application is about

Feedback Form: <https://forms.gle/PJz7v57D9GJUCw4A>

Changes Made Based on Feedback

- Added glassmorphism overlay on hero sections for better text readability (User 2)
- Enhanced homepage with statistics preview cards (User 5)

- Improved hero description clarity (User 5)
- Adjusted text contrast for better readability (User 1, 5)

D. Development Process: Developing the Code

HTML Structure

All 5 HTML pages follow semantic HTML5 structure: header, nav, main, section, footer, article, time.

CSS Implementation

Features include: CSS variables for theming, three breakpoints (1024px, 768px, 480px), glassmorphism, WCAG AA contrast, reduced motion support, flexbox and grid layouts.

JavaScript Architecture

- app.js – Main orchestrator: fetches JSON, validates, renders
- renderer.js – Custom template engine: JSON to HTML
- router.js – Navigation: page detection, hamburger toggle
- validator.js – Data validation: fields, types, ranges, URLs

Template Engine

The custom template engine converts JSON to HTML strings with XSS protection via `escapeHTML()`. This approach was chosen for simplicity, maintainability, and GitHub Pages compatibility.

JSON Validation

validator.js validates all JSON data before rendering:

- Required fields (year, title, category, description, significance, source, page)
- Data types (year must be number, URLs valid)
- Year range (1980–2026)
- Minimum string lengths (10 characters)
- Valid categories and pages

Data Flow

Browser → app.js fetch('data/timeline.json') → validator.js → renderer.js → DOM

Responsive Implementation

- 1024px and above – Multi-column, horizontal nav, grid cards
- 768–1023px – 2-column grid, responsive spacing
- Below 768px – Single-column, hamburger nav, stacked cards

E. Testing: Code Validation Reports and Actions Taken

HTML Validation (W3C Nu HTML Checker)

All 5 pages were validated using the W3C Nu HTML Checker. The pages passed validation with 0 errors and 0 warnings on the first attempt.

Page	Result
index.html	✓ 0 errors
origins.html	✓ 0 errors
growth.html	✓ 0 errors
smart-nation.html	✓ 0 errors
timeline.html	✓ 0 errors

Live Validation: <https://validator.w3.org/nu/?doc=https://varungiri970-jpg.github.io/singapore-connected/index.html>

Accessibility Testing (Lighthouse)

All 5 pages – Desktop Score: 90/100 – Mobile Score: 90/100

Issues: Minor contrast warnings. Fix: Changed p colour from #3A2A2A to #1A0000 and added underline on hover for links.

Lighthouse Full Reports

Desktop (Average): Performance 89/100, Accessibility 90/100, Best Practices 100/100, SEO 100/100

Mobile (Average): Performance 68/100, Accessibility 90/100, Best Practices 100/100, SEO 100/100

Reports: documents/Report/Lighthouse-Report-Viewer-PC.pdf and ...-Mobile.pdf

Responsive Testing

- Desktop (1440px) – Full multi-column – Passed
- Laptop (1024px) – Multi-column – Passed
- Tablet (768px) – 2-column grid – Passed
- Mobile (375px) – Single-column, hamburger – Passed

Console & Deployment Testing

All pages loaded with 0 console errors. JSON data loads successfully via fetch(). GitHub Pages deployment passes all tests.

Live URL: <https://varungiri970-jpg.github.io/singapore-connected/>

Testing Summary

Test	Tool	Result
HTML Validation	W3C Nu HTML Checker	✓ 0 errors
Accessibility	Google Lighthouse	✓ 90/100
Best Practices	Google Lighthouse	✓ 100/100
SEO	Google Lighthouse	✓ 100/100
Responsive Design	Chrome DevTools	✓ Passed
Console Errors	Browser DevTools	✓ None
JSON Validation	Custom validator	✓ Passed
GitHub Pages	Live URL	✓ Working

F. Reflections on What I Learned

Key Learning Outcomes

- 1. Research is Foundation: Verifying historical claims with primary sources (GovTech, IMDA, Smart Nation) ensured accuracy. Every milestone has a verifiable source URL.
- 2. Dynamic Content with JSON: Using a template engine to render content from JSON makes websites more maintainable and scalable.
- 3. JSON Validation: Validating data before rendering prevents runtime errors and ensures data integrity.
- 4. Accessibility is Essential: Semantic HTML, ARIA labels, and proper contrast are fundamental for inclusive digital experiences.
- 5. Responsive Design: Mobile-first thinking ensures websites work on all devices.
- 6. Project Management: Phased development (research → planning → development → testing → submission) made the project manageable.
- 7. Premium Design: Consistent colours (Singapore flag red and white), typography, and glassmorphism create a polished appearance.
- 8. GitHub Pages: Static sites with relative paths can be deployed for free without a backend server.

Challenges & Solutions

Challenge	Solution
Hero images not loading on GitHub Pages	Changed /images/ to images/ (relative paths)
Heading hierarchy validation	Changed h4 to h3 in footer
Contrast accessibility warning	Changed #3A2A2A to #1A0000 for WCAG AA compliance
JSON data validation	Created comprehensive validator checking all required fields
Homepage clarity	Added statistics preview and improved hero content
Mobile hamburger not closing on link click	Added click handler to close menu on navigation

Future Improvements

- Add a REST API for dynamic data updates
- Allow user contributions to the timeline
- Add more interactive visualisations
- Implement dark/light theme toggle

Conclusion

This project successfully demonstrated all CM1040 coursework requirements: research with verified sources, 5-page responsive website, custom template engine with JSON, JSON validation, HTML validation (0 errors), accessibility (90/100), GitHub Pages deployment, and comprehensive testing evidence.

Important Links

Live Demo: <https://varungiri970-jpg.github.io/singapore-connected/>

GitHub Repository: <https://github.com/varungiri970-jpg/singapore-connected>

HTML Validation: <https://validator.w3.org/nu/?doc=https://varungiri970-jpg.github.io/singapore-connected/index.html>

Visily Wireframes: <https://app.visily.ai/projects/24a26e2d-5b8c-4ee6-b74d-65b0fcf52f70/boards/2697893>

Feedback Form: <https://forms.gle/PJz7v57D9GJXUCw4A>